ROCSCIENCE

Case12_MaxOutput
SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

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Slide2 Analysis Information

Case12_MaxOutput

Project Summary

File Name:	Case12_MaxOutput.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:07.669s
Project Title:	SLIDE - An Interactive Slope Stability Program
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	50
Tolerance:	0.005
Maximum number of iterations:	75
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

top layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	29.4 kPa
Phi	12 °
Water Surface	None
Ru Value	0

middle layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	9.8 kPa
Phi	5 °
Water Surface	None
Ru Value	0

lower layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	294 kPa
Phi	40 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	0.420185
Center:	26.967, 45.140
Radius:	31.450
Left Slip Surface Endpoint:	17.987, 15.000
Right Slip Surface Endpoint:	56.737, 35.000
Resisting Moment:	34669.4 kN-m
Driving Moment:	82509.8 kN-m
Total Slice Area:	350.585 m2
Surface Horizontal Width:	38.7497 m
Surface Average Height:	9.04742 m

Method: spencer

FS	0.418145
Center:	26.126, 44.854
Radius:	30.942
Left Slip Surface Endpoint:	17.995, 15.000
Right Slip Surface Endpoint:	55.457, 35.000
Resisting Moment:	32052.9 kN-m
Driving Moment:	76654.8 kN-m
Resisting Horizontal Force:	843.493 kN
Driving Horizontal Force:	2017.22 kN
Total Slice Area:	320.546 m2
Surface Horizontal Width:	37.4616 m
Surface Average Height:	8.55666 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	8089
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	5077
Number of Invalid Surfaces:	3012

Error Codes

Error Code -108 reported for 102 surfaces
Error Code -111 reported for 2910 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.420185

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.781771	5.34316	-15.8501	middle layer	9.8	5	26.2917	11.0474	14.2575	0	14.2575	6.79282	6.79282
2	0.781771	16.117	-14.3747	middle layer	9.8	5	29.1631	12.2539	28.0482	0	28.0482	20.5741	20.5741
3	0.781771	26.5771	-12.9089	middle layer	9.8	5	31.9161	13.4107	41.2699	0	41.2699	33.955	33.955
4	0.781771	36.7283	-11.4517	middle layer	9.8	5	34.5545	14.5193	53.9415	0	53.9415	46.9417	46.9417
5	0.781771	46.5757	-10.0019	middle layer	9.8	5	37.082	15.5813	66.0803	0	66.0803	59.5404	59.5404
6	0.781771	56.1236	-8.55862	middle layer	9.8	5	39.5017	16.598	77.7019	0	77.7019	71.757	71.757
7	0.781771	65.3757	-7.12078	middle layer	9.8	5	41.8166	17.5707	88.8197	0	88.8197	83.5958	83.5958
8	0.781771	74.335	-5.68743	middle layer	9.8	5	44.0291	18.5004	99.4457	0	99.4457	95.0608	95.0608
9	0.781771	83.004	-4.25764	middle layer	9.8	5	46.1413	19.3879	109.59	0	109.59	106.155	106.155
10	0.781771	91.3846	-2.83051	middle layer	9.8	5	48.1552	20.2341	119.262	0	119.262	116.881	116.881
11	0.781771	99.4781	-1.40514	middle layer	9.8	5	50.072	21.0395	128.468	0	128.468	127.24	127.24
12	0.781771	107.285	0.0193669	middle layer	9.8	5	51.8936	21.8049	137.217	0	137.217	137.234	137.234
13	0.781771	114.807	1.44388	middle layer	9.8	5	53.6204	22.5305	145.51	0	145.51	146.862	146.862
14	0.781771	122.041	2.86929	middle layer	9.8	5	55.2538	23.2168	153.355	0	153.355	156.124	156.124
15	0.781771	128.989	4.29648	middle layer	9.8	5	56.7943	23.8641	160.753	0	160.753	165.02	165.02
16	0.781771	135.649	5.72635	middle layer	9.8	5	58.242	24.4724	167.707	0	167.707	173.547	173.547
17	0.781771	142.017	7.15981	middle layer	9.8	5	59.5976	25.042	174.217	0	174.217	181.703	181.703
18	0.781771	148.094	8.59779	middle layer	9.8	5	60.8606	25.5727	180.283	0	180.283	189.485	189.485
19	0.781771	153.874	10.0413	middle layer	9.8	5	62.0312	26.0646	185.905	0	185.905	196.889	196.889
20	0.781771	159.354	11.4912	middle layer	9.8	5	63.1086	26.5173	191.081	0	191.081	203.91	203.91
21	0.781771	164.531	12.9486	middle layer	9.8	5	64.0927	26.9308	195.805	0	195.805	210.542	210.542
22	0.781771	169.399	14.4147	middle layer	9.8	5	64.9821	27.3045	200.078	0	200.078	216.78	216.78
23	0.781771	173.951	15.8904	middle layer	9.8	5	65.7758	27.638	203.89	0	203.89	222.614	222.614
24	0.781771	178.183	17.377	middle layer	9.8	5	66.4726	27.9308	207.236	0	207.236	228.038	228.038
25	0.781771	182.085	18.8759	middle layer	9.8	5	67.0709	28.1822	210.109	0	210.109	233.041	233.041
26	0.781771	185.649	20.3882	middle layer	9.8	5	67.5688	28.3914	212.5	0	212.5	237.613	237.613
27	0.781771	188.866	21.9156	middle layer	9.8	5	67.9639	28.5574	214.398	0	214.398	241.741	241.741
28	0.781771	191.724	23.4595	middle layer	9.8	5	68.2535	28.6791	215.79	0	215.79	245.41	245.41
29	0.781771	194.212	25.0218	middle layer	9.8	5	68.4351	28.7554	216.662	0	216.662	248.605	248.605
30	0.781771	196.315	26.6041	middle layer	9.8	5	68.5051	28.7848	216.998	0	216.998	251.309	251.309

31	0.781771	198.018	28.2087	middle layer	9.8	5	68.4596	28.7657	216.779	0	216.779	253.5	253.5
32	0.781771	199.303	29.8379	middle layer	9.8	5	68.2942	28.6962	215.985	0	215.985	255.157	255.157
33	0.781771	200.149	31.494	middle layer	9.8	5	68.0039	28.5742	214.591	0	214.591	256.254	256.254
34	0.781771	200.533	33.1801	middle layer	9.8	5	67.5826	28.3972	212.567	0	212.567	256.759	256.759
35	0.781771	200.429	34.8993	middle layer	9.8	5	67.024	28.1625	209.885	0	209.885	256.64	256.64
36	0.781771	199.805	36.6553	middle layer	9.8	5	66.3201	27.8667	206.503	0	206.503	255.856	255.856
37	0.781771	198.627	38.4524	middle layer	9.8	5	65.4619	27.5061	202.382	0	202.382	254.364	254.364
38	0.781771	196.851	40.2955	middle layer	9.8	5	64.439	27.0763	197.468	0	197.468	252.108	252.108
39	0.781771	193.006	42.1904	middle layer	9.8	5	62.9194	26.4378	190.171	0	190.171	247.204	247.204
40	0.781771	182.797	44.144	middle layer	9.8	5	59.9598	25.1942	175.956	0	175.956	234.15	234.15
41	0.781771	171.225	46.1647	middle layer	9.8	5	56.7	23.8245	160.3	0	160.3	219.354	219.354
42	0.781771	158.789	48.2627	middle layer	9.8	5	53.2551	22.377	143.756	0	143.756	203.45	203.45
43	0.781771	145.378	50.451	middle layer	9.8	5	49.6051	20.8433	126.226	0	126.226	186.297	186.297
44	0.781771	130.852	52.7459	middle layer	9.8	5	45.7255	19.2131	107.593	0	107.593	167.716	167.716
45	0.781771	115.025	55.1693	middle layer	9.8	5	41.5845	17.4732	87.705	0	87.705	147.469	147.469
46	0.781771	97.6441	57.7509	middle layer	9.8	5	37.1411	15.6061	66.3643	0	66.3643	125.231	125.231
47	0.697054	70.8967	60.37	top layer	29.4	12	64.4341	27.0742	-10.9418	0	-10.9418	102.345	102.345
48	0.697054	53.863	63.0559	top layer	29.4	12	54.8135	23.0318	-29.96	0	-29.96	77.8774	77.8774
49	0.697054	34.5891	66.0198	top layer	29.4	12	44.6161	18.747	-50.1183	0	-50.1183	50.1844	50.1844
50	0.697054	12.1551	69.3861	top layer	29.4	12	33.7102	14.1645	-71.6775	0	-71.6775	17.941	17.941

Global Minimum Query (spencer) - Safety Factor: 0.418145

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.747159	4.81832	-14.5207	middle layer	9.8	5	26.2064	10.9581	13.2368	0	13.2368	6.44921	6.44921
2	0.747159	14.4049	-13.0958	middle layer	9.8	5	28.8763	12.0745	25.9974	0	25.9974	19.2799	19.2799
3	0.747159	23.717	-11.6792	middle layer	9.8	5	31.4382	13.1457	38.2418	0	38.2418	31.7432	31.7432
4	0.747159	32.7588	-10.2697	middle layer	9.8	5	33.8955	14.1732	49.986	0	49.986	43.8447	43.8447
5	0.747159	41.5341	-8.8665	middle layer	9.8	5	36.2511	15.1582	61.2449	0	61.2449	55.5898	55.5898
6	0.747159	50.0464	-7.46864	middle layer	9.8	5	38.5079	16.1019	72.0307	0	72.0307	66.9825	66.9825
7	0.747159	58.2983	-6.07524	middle layer	9.8	5	40.6681	17.0052	82.3552	0	82.3552	78.0268	78.0268
8	0.747159	66.2921	-4.68544	middle layer	9.8	5	42.7338	17.8689	92.2282	0	92.2282	88.7258	88.7258
9	0.747159	74.0295	-3.2984	middle layer	9.8	5	44.7068	18.6939	101.658	0	101.658	99.0814	99.0814
10	0.747159	81.5119	-1.9133	middle layer	9.8	5	46.5886	19.4808	110.652	0	110.652	109.096	109.096
11	0.747159	88.74	-0.52931	middle layer	9.8	5	48.3806	20.2301	119.217	0	119.217	118.77	118.77
12	0.747159	95.7144	0.854368	middle layer	9.8	5	50.0838	20.9423	127.357	0	127.357	128.104	128.104
13	0.747159	102.435	2.23855	middle layer	9.8	5	51.6993	21.6178	135.078	0	135.078	137.099	137.099
14	0.747159	108.901	3.62403	middle layer	9.8	5	53.2275	22.2568	142.382	0	142.382	145.753	145.753
15	0.747159	115.112	5.01164	middle layer	9.8	5	54.6691	22.8596	149.272	0	149.272	154.066	154.066
16	0.747159	121.066	6.40221	middle layer	9.8	5	56.0241	23.4262	155.748	0	155.748	162.035	162.035
17	0.747159	126.761	7.79658	middle layer	9.8	5	57.2928	23.9567	161.813	0	161.813	169.657	169.657
18	0.747159	132.196	9.19561	middle layer	9.8	5	58.4754	24.4512	167.464	0	167.464	176.93	176.93
19	0.747159	137.366	10.6002	middle layer	9.8	5	59.5714	24.9095	172.702	0	172.702	183.85	183.85
20	0.747159	142.27	12.0113	middle layer	9.8	5	60.5802	25.3313	177.525	0	177.525	190.414	190.414
21	0.747159	146.902	13.4298	middle layer	9.8	5	61.5016	25.7166	181.927	0	181.927	196.613	196.613
22	0.747159	151.258	14.8567	middle layer	9.8	5	62.3344	26.0648	185.908	0	185.908	202.443	202.443
23	0.747159	155.333	16.2932	middle layer	9.8	5	63.0776	26.3756	189.46	0	189.46	207.897	207.897
24	0.747159	159.121	17.7402	middle layer	9.8	5	63.7303	26.6485	192.579	0	192.579	212.967	212.967
25	0.747159	162.616	19.1991	middle layer	9.8	5	64.2906	26.8828	195.257	0	195.257	217.644	217.644
26	0.747159	165.809	20.671	middle layer	9.8	5	64.7567	27.0777	197.486	0	197.486	221.918	221.918
27	0.747159	168.692	22.1573	middle layer	9.8	5	65.1269	27.2325	199.255	0	199.255	225.776	225.776
28	0.747159	171.255	23.6596	middle layer	9.8	5	65.3988	27.3462	200.553	0	200.553	229.206	229.206
29	0.747159	173.488	25.1793	middle layer	9.8	5	65.5696	27.4176	201.369	0	201.369	232.195	232.195
30	0.747159	175.379	26.7182	middle layer	9.8	5	65.6361	27.4454	201.688	0	201.688	234.726	234.726
31	0.747159	176.913	28.2782	middle layer	9.8	5	65.5949	27.4282	201.491	0	201.491	236.778	236.778
32	0.747159	178.075	29.8614	middle layer	9.8	5	65.4421	27.3643	200.761	0	200.761	238.334	238.334
33	0.747159	178.848	31.4702	middle layer	9.8	5	65.1733	27.2519	199.476	0	199.476	239.368	239.368

34	0.747159	179.211	33.1072	middle layer	9.8	5	64.783	27.0887	197.611	0	197.611	239.854	239.854
35	0.747159	179.142	34.7753	middle layer	9.8	5	64.2655	26.8723	195.137	0	195.137	239.762	239.762
36	0.747159	178.615	36.4779	middle layer	9.8	5	63.6138	26.5998	192.022	0	192.022	239.055	239.055
37	0.747159	177.598	38.2187	middle layer	9.8	5	62.8198	26.2678	188.228	0	188.228	237.695	237.695
38	0.747159	176.058	40.0024	middle layer	9.8	5	61.8745	25.8725	183.71	0	183.71	235.633	235.633
39	0.747159	173.951	41.834	middle layer	9.8	5	60.767	25.4094	178.416	0	178.416	232.813	232.813
40	0.747159	171.229	43.7197	middle layer	9.8	5	59.4846	24.8732	172.287	0	172.287	229.171	229.171
41	0.747159	165.353	45.6668	middle layer	9.8	5	57.4396	24.0181	162.513	0	162.513	221.305	221.305
42	0.747159	154.294	47.6844	middle layer	9.8	5	54.1901	22.6593	146.983	0	146.983	206.504	206.504
43	0.747159	142.312	49.7835	middle layer	9.8	5	50.7343	21.2143	130.466	0	130.466	190.467	190.467
44	0.747159	129.381	51.978	middle layer	9.8	5	47.0713	19.6826	112.959	0	112.959	173.16	173.16
45	0.747159	115.356	54.2863	middle layer	9.8	5	43.1746	18.0533	94.3355	0	94.3355	154.389	154.389
46	0.747159	100.042	56.7327	middle layer	9.8	5	39.0102	16.3119	74.4314	0	74.4314	133.893	133.893
47	0.773067	85.7168	59.4001	top layer	29.4	12	68.119	28.4836	-4.31128	0	-4.31128	110.872	110.872
48	0.773067	65.4737	62.3488	top layer	29.4	12	57.536	24.0584	-25.1302	0	-25.1302	84.6871	84.6871
49	0.773067	42.3275	65.6259	top layer	29.4	12	46.2497	19.3391	-47.3329	0	-47.3329	54.7469	54.7469
50	0.773067	14.9576	69.3948	top layer	29.4	12	34.0741	14.2479	-71.2846	0	-71.2846	19.343	19.343

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.420185

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	17.9873	15	0	0	0
2	18.7691	14.778	23.6033	0	0
3	19.5508	14.5777	51.8939	0	0
4	20.3326	14.3985	84.0996	0	0
5	21.1144	14.2401	119.504	0	0
6	21.8961	14.1023	157.442	0	0
7	22.6779	13.9846	197.292	0	0
8	23.4597	13.887	238.474	0	0
9	24.2415	13.8091	280.444	0	0
10	25.0232	13.7509	322.692	0	0
11	25.805	13.7122	364.737	0	0
12	26.5868	13.6931	406.126	0	0
13	27.3685	13.6933	446.43	0	0
14	28.1503	13.713	485.247	0	0
15	28.9321	13.7522	522.192	0	0
16	29.7139	13.8109	556.901	0	0
17	30.4956	13.8893	589.03	0	0
18	31.2774	13.9875	618.252	0	0
19	32.0592	14.1057	644.254	0	0
20	32.8409	14.2442	666.742	0	0
21	33.6227	14.4031	685.434	0	0
22	34.4045	14.5829	700.063	0	0
23	35.1863	14.7838	710.376	0	0
24	35.968	15.0063	716.133	0	0
25	36.7498	15.251	717.108	0	0
26	37.5316	15.5183	713.087	0	0
27	38.3133	15.8088	703.871	0	0
28	39.0951	16.1234	689.273	0	0
29	39.8769	16.4626	669.122	0	0
30	40.6586	16.8275	643.261	0	0
31	41.4404	17.2191	611.55	0	0
32	42.2222	17.6384	573.866	0	0
33	43.004	18.0868	530.107	0	0
34	43.7857	18.5658	480.193	0	0
35	44.5675	19.077	424.068	0	0
36	45.3493	19.6223	361.71	0	0
37	46.131	20.2041	293.13	0	0
38	46.9128	20.8249	218.382	0	0
39	47.6946	21.4878	137.577	0	0
40	48.4764	22.1964	51.7293	0	0
41	49.2581	22.9552	-35.1663	0	0
42	50.0399	23.7694	-121.608	0	0
43	50.8217	24.6457	-206.18	0	0
44	51.6034	25.5924	-287.118	0	0
45	52.3852	26.6203	-362.169	0	0
46	53.167	27.7438	-428.382	0	0
47	53.9488	28.9829	-481.739	0	0
48	54.6458	30.2085	-423.668	0	0
49	55.3429	31.5798	-344.588	0	0
50	56.0399	33.1469	-235.124	0	0
51	56.737	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 0.418145

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	17.9953	15	0	0	0
2	18.7424	14.8065	22.143	0	0
3	19.4896	14.6327	48.2381	0	0
4	20.2367	14.4782	77.6351	0	0
5	20.9839	14.3429	109.729	0	0
6	21.7311	14.2263	143.954	0	0
7	22.4782	14.1283	179.783	0	0
8	23.2254	14.0488	216.719	0	0
9	23.9725	13.9876	254.297	0	0
10	24.7197	13.9445	292.08	0	0
11	25.4669	13.9196	329.653	0	0
12	26.214	13.9127	366.626	0	0
13	26.9612	13.9238	402.629	0	0
14	27.7083	13.953	437.314	0	0
15	28.4555	14.0003	470.348	0	0
16	29.2026	14.0659	501.416	0	0
17	29.9498	14.1497	530.22	0	0
18	30.697	14.252	556.476	0	0
19	31.4441	14.373	579.913	0	0
20	32.1913	14.5128	600.276	0	0
21	32.9384	14.6717	617.322	0	0
22	33.6856	14.8502	630.818	0	0
23	34.4328	15.0484	640.548	0	0
24	35.1799	15.2667	646.304	0	0
25	35.9271	15.5058	647.891	0	0
26	36.6742	15.7659	645.129	0	0
27	37.4214	16.0478	637.845	0	0
28	38.1686	16.3521	625.882	0	0
29	38.9157	16.6795	609.097	0	0
30	39.6629	17.0307	587.358	0	0
31	40.41	17.4068	560.551	0	0
32	41.1572	17.8087	528.577	0	0
33	41.9044	18.2377	491.356	0	0
34	42.6515	18.695	448.828	0	0
35	43.3987	19.1822	400.957	0	0
36	44.1458	19.701	347.737	0	0
37	44.893	20.2535	289.192	0	0
38	45.6402	20.8418	225.387	0	0
39	46.3873	21.4688	156.435	0	0
40	47.1345	22.1376	82.509	0	0
41	47.8816	22.8521	3.85835	0	0
42	48.6288	23.6169	-77.5057	0	0
43	49.3759	24.4376	-157.639	0	0
44	50.1231	25.3212	-235.013	0	0
45	50.8703	26.2767	-307.781	0	0
46	51.6174	27.316	-373.56	0	0
47	52.3646	28.4549	-429.178	0	0
48	53.1377	29.7621	-370.879	0	0
49	53.9107	31.2376	-289.316	0	0
50	54.6838	32.9439	-172.798	0	0
51	55.4569	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 3 96, 3 96, 35 72, 35 48, 35 24, 19 18, 15 0, 15
Material Boundary	24, 19 72, 35
Material Boundary	0, 3 96, 35